

Definition 26. A continuous random variable X having the bell shaped distribution is called normal random variable. The density of the normal random variable X with mean μ and variance σ^2 is $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \textcircled{1} \quad f(x) \geq 0$$

$$\textcircled{2} \quad \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left[\frac{x-\mu}{\sigma} \right]^2} dx \quad \text{Let } t = \frac{x-\mu}{\sigma} \\ dt = dx$$

$$\boxed{\int_0^{\infty} e^{-t^2/2} dt = \sqrt{\frac{\pi}{2}}}$$

$$\begin{aligned} &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} t^2} dt \quad \mu=0, \sigma=1 \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} t^2} dt \quad N(0,1) \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t^2/2} dt \\ &= \sqrt{\frac{2}{\pi}} \sqrt{\frac{\pi}{2}} = 1 \end{aligned}$$

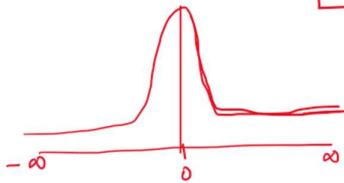
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

The fun $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \left[\frac{x-\mu}{\sigma} \right]^2}$ is pdf.

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \left[\frac{x-\mu}{\sigma} \right]^2} dx$$

$$\boxed{E(x) = \mu}$$

$$\boxed{S.D = \sigma}$$



$$\mu=0, \sigma=1$$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2}$$

$$P(-\infty < x < \infty) = 1$$

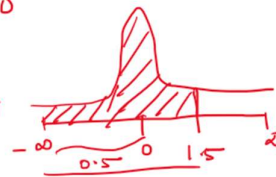
$$P(-\infty < x < 0) + P(0 < x < \infty) = 1$$

$$P(-\infty < x < 0) = 0.5, \quad P(0 < x < \infty) = 0.5$$

$$P(-\infty < x < 1.05) = P(-\infty < x < 0) + P(0 < x < 1.05)$$

$$= 0.5 + P(0 < x < 1.05)$$

$$P(0 < x < 1.05) = 0.3531 = \int_0^{1.05} f(x) dx$$

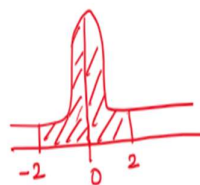


$$\int_0^{1.05} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} dx$$

$$P(-\infty < x < 1.05) = 0.5 + 0.3531$$

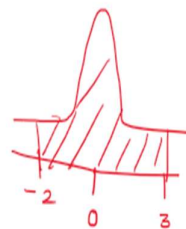
$$\textcircled{*} \quad \text{Find } P(-\infty < x < 2.25) = P(-\infty < x < 0) + P(0 < x < 2.25) \\ = 0.5 + 0.4878$$

⑧ Find $P(-2 < x < 2) = P(0 < x < 2)$
 $= 0.4772$.

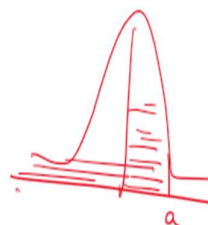
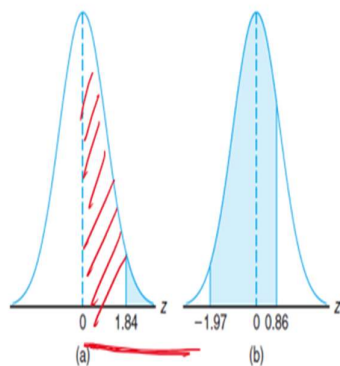


⑨ Find $P(-2 < x < 3)$.

$$\begin{aligned} P(-2 < x < 3) &= P(-2 < x < 0) + P(0 < x < 3) \\ &= P(0 < x < 2) + P(0 < x < 3) \\ &= 0.4772 + 0.4987 \end{aligned}$$



Example 58. Given a standard normal distribution, find the area under the curve that lies to the right of $z = 1.84$ and between $z = -1.97, z = 0.86$



$$P(0 < z < a)$$

$$\begin{aligned} P(z \geq 1.84) &= 0.5 - P(0 < z < 1.84) \\ &= 0.5 - 0.4671 \end{aligned}$$

$$\begin{aligned} P(-1.97 < z < 0.86) &= P(-1.97 < z < 0) + P(0 < z < 0.86) \\ &= P(0 < z < 1.97) + P(0 < z < 0.86) \end{aligned}$$

$$= 0.4756 + 0.3051$$

Example 59. Given standard normal distribution, find the values of k such that

(a) $P(Z > k) = 0.3015$

(b) $P(k < Z < -0.18) = 0.4197$

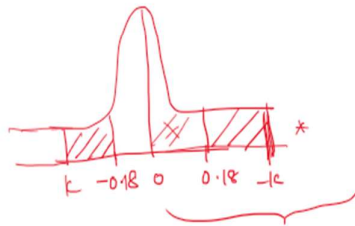
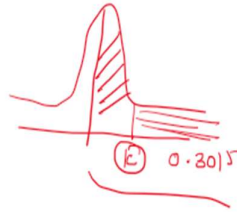
$$P(Z > k) = 0.3015$$

$$P(0 < Z < k) = 0.5 - P(Z > k)$$

$$= 0.5 - 0.3015$$

$$= 0.19$$

$$k = 0.52$$



$$P(k < Z < -0.18)$$

$$= P(0.18 < Z < -k)$$

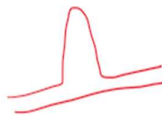
$$0.4197 = 0.5 - (P(0 < Z < 0.18) + P(Z > -k))$$

$$0.4197 = 0.5 - 0.0714 + P(Z > -k)$$

$$P(Z > -k) = 0.4197 - 0.5 + 0.0714$$

$$k = -2.37$$

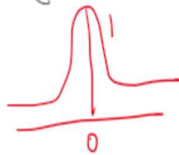
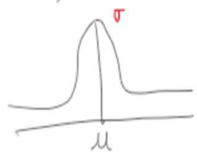
Normal dist.: $\mu = 0, \sigma = 1$.



$$f(x) = k e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \sim N(\mu, \sigma) \quad \text{Normal dist}$$

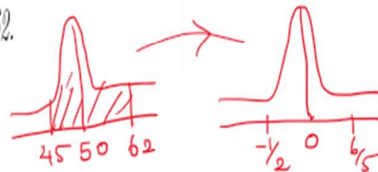
$$f(x) = k e^{-\frac{1}{2}x^2} \sim N(0, 1) \quad \text{std Normal dist.}$$

$N(\mu, \sigma) \sim N(0, 1)$ by finding area



Example 60. Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$. Find the probability that X assumes a value between 45 and 62. $z = \frac{x-45}{10}$

$$z = \frac{x-\mu}{\sigma} \sim N(0, 1)$$



$$z = \frac{x-50}{10}, \quad \text{if } x=45 \text{ then } z = \frac{45-50}{10} = \frac{-5}{10} = -\frac{1}{2}$$

$$\text{if } x=62, \text{ then } z = \frac{62-50}{10} = \frac{12}{10} = \frac{6}{5}$$

$$P(45 < X < 62) ?$$

$$P(45 < X < 62) = P(-\frac{1}{2} < Z < \frac{6}{5}) = P(-\frac{1}{2} < Z < 0) + P(0 < Z < \frac{6}{5})$$

$$= P(0 < Z < \frac{1}{2}) + P(0 < Z < \frac{6}{5})$$

Example 61. The weekly wages of 1000 workers are normally distributed around a mean of Rs.70 with a standard deviation of Rs.5. Estimate the number of workers whose weekly wages will be

- between Rs.69 and Rs.72
- less than Rs.69
- more than Rs.72

Example 62. An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

$$\mu = 800, \sigma = 40, z = \frac{x - \mu}{\sigma}$$

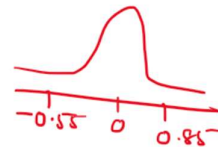
$$P(778 \leq X \leq 834) = P\left(\frac{778 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{834 - \mu}{\sigma}\right)$$

$$= P\left(\frac{778 - 800}{40} < Z < \frac{834 - 800}{40}\right)$$

$$= P(-0.55 < Z < 0.85)$$

$$= P(-0.55 < Z < 0) + P(0 < Z \leq 0.85)$$

$$= P(0 < Z < 0.55) + P(0 < Z \leq 0.85)$$



Binomial dist: $\text{mean} = np, \text{var} = npq$

$$z = \frac{x - \mu}{\sigma} = \frac{x - np}{\sqrt{npq}}$$